

PeptiVerse: A unified platform for therapeutic peptide property prediction

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Therapeutic peptides combine the advantages of small molecules and antibodies, offering target flexibility and low immunogenicity, yet their successful translation requires careful evaluation of multiple developability properties beyond binding alone. As chemically modified peptides become increasingly common in drug design, no unified platform currently supports systematic property assessment across both canonical sequences and SMILES-based representations. Leveraging the generalizability of large foundational models trained on protein and chemical data, we introduce PeptiVerse, a universal therapeutic peptide property prediction platform. PeptiVerse accepts either amino acid sequences or chemically modified peptide SMILES, delivers state-of-the-art performance across diverse property prediction tasks, and provides both a web interface and open-source implementation for rapid, accessible, and scalable peptide developability analysis. By unifying property prediction across representations, PeptiVerse directly supports early-stage peptide therapeutic development campaigns and property-aware generative design workflows.

Peptide-based therapeutics have gained significant attention in recent years, highlighted by the clinical success of GLP-1 receptor agonists for metabolic diseases^{1,2}. As a therapeutic modality, peptides occupy a unique position between small molecules and antibodies, combining larger interaction surfaces capable of engaging protein-protein interfaces traditionally considered undruggable^{3,4} with reduced immunogenicity and manufacturing complexity relative to full-length antibodies^{4–6}. These features make peptides candidates for a broad spectrum of therapeutic targets, including receptors, enzymes, and intrinsically disordered proteins.

Despite these advantages, native peptides often exhibit sub-optimal translational profiles. For example, poor membrane permeability limits cellular uptake and oral bioavailability^{7,8}, rapid proteolytic degradation results in short circulating half-life and frequent dosing^{4,9}, and low solubility promotes aggregation and formulation failure^{10,11}. In addition, certain features of amphipathic sequences can cause hemolysis through nonspecific membrane disruption^{12,13}, while nonspecific protein adsorption (fouling) reduces effective concentration and

increases off-target interactions in complex biological environments^{14,15}. These limitations can be partially mitigated through cyclization, terminal modifications, D-amino acids, or other noncanonical residues^{4,16,17}, but such interventions push peptides beyond the assumptions of traditional protein sequence-based predictors^{18,19}. As a result, practical peptide design requires systematic evaluation of multiple experimentally grounded properties beyond binding affinity alone.

Existing computational tools inadequately address this need. Sequence-based predictors such as PeptideBERT and those in Peptipedia v2.0 are confined to natural amino acids and cannot accommodate chemical modifications^{15,20}. Small-molecule ADMET platforms accept “chemical” inputs in the form of SMILES but are trained on drug-like chemical space that differs substantially from peptides and proteins^{21,22}. Peptide-oriented SMILES predictors, including PepLand, PepDoRA, and PeptideDashboard, represent important steps toward chemistry-aware peptide modeling^{23–25}, yet cover only a limited subset of relevant properties. Fitness-focused platforms such as ProteinGym benchmark mutational effects but do not target critical biochemical

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traits relevant for therapeutic peptide design^{18,19}. Together, these gaps highlight the lack of a comprehensive, modality-flexible predictor capable of evaluating the full chemical and sequence diversity of modern peptide therapeutics.

The recent emergence of large protein and chemical language models enables unified property predictors that leverage rich learned representations without requiring explicit structural information^{26,27}. Such predictors have become integral to modern generative peptide design workflows, where classifiers guide sampling, rank candidates, or provide post hoc filtering^{28–36}. In this setting, gradient coupling between generator and predictor is not always necessary, allowing effective use of classical methods such as SVM³⁷, Elastic Nets³⁸, and XGBoost³⁹, which perform well on pretrained embeddings while reducing the risk of overfitting⁴⁰.

To realize this potential in a unified framework, we introduce PeptiVerse (Fig. 1), a universal therapeutic peptide property evaluation platform developed to standardize and accelerate computational peptide design. PeptiVerse integrates state-of-the-art foundation models with carefully curated datasets to deliver fast, accurate, and scalable property predictions, supporting both sequence-based and SMILES-based peptide inputs. Beyond post hoc evaluation, these predictors can be directly used as guidance oracles within generative modeling workflows^{29,30,32–34}, enabling generation, ranking, and optimization of peptide candidates across diverse targets. Together, PeptiVerse provides a unified foundation for property-aware peptide discovery, enabling both early-stage candidate prioritization and integration with generative design workflows to accelerate therapeutic translation.

Results

Dataset composition highlights constraints across peptide properties

Accurate peptide property prediction is fundamentally constrained by data availability, coverage, and experimental diversity. As an initial

step, we curated and integrated experimentally derived datasets spanning multiple peptide properties from a wide range of public sources^{15,16,23,41–46}. These datasets collectively cover both canonical amino acid sequences and chemically modified peptides represented as SMILES, enabling unified analysis across representation modalities. Examination of dataset composition reveals substantial heterogeneity in dataset size, label balance, and value distributions across properties (Fig. 2 and Table 2).

Our data show that several classification tasks are supported by large, well-populated datasets with broad coverage of peptide sequence and chemical space. Hemolysis and non-fouling datasets comprise thousands to tens of thousands of peptides curated from antimicrobial and surface-interaction studies^{41,47}, while solubility datasets aggregate protein expression outcomes from structural genomics pipelines and mutational databases^{43,48}. Permeability datasets include both canonical cell-penetrating peptides and non-canonical cyclic peptides measured using PAMPA and Caco-2 assays^{16,49,50}, yielding relatively balanced class distributions and continuous-valued measurements spanning multiple orders of magnitude. These properties exhibit broad value distributions and sufficient sample sizes to support robust model training and evaluation under similarity-aware splits.

In contrast, regression tasks such as peptide half-life and binding affinity remain comparatively data-limited. Half-life measurements, curated from THPdb2, PEPlife, and PepTherDia^{44–46}, are sparse, heterogeneous in experimental protocol, and often reported in coarse or qualitative units, resulting in limited sample sizes for both sequence- and SMILES-based representations. Binding affinity datasets aggregate diverse experimental readouts (K_d , K_i , and IC_{50}) across protein-peptide pairs^{23,51}, but remain modest in scale relative to classification tasks. Together, our observations highlight that achievable predictive performance across peptide properties is frequently constrained by data availability and experimental variability rather than model capacity

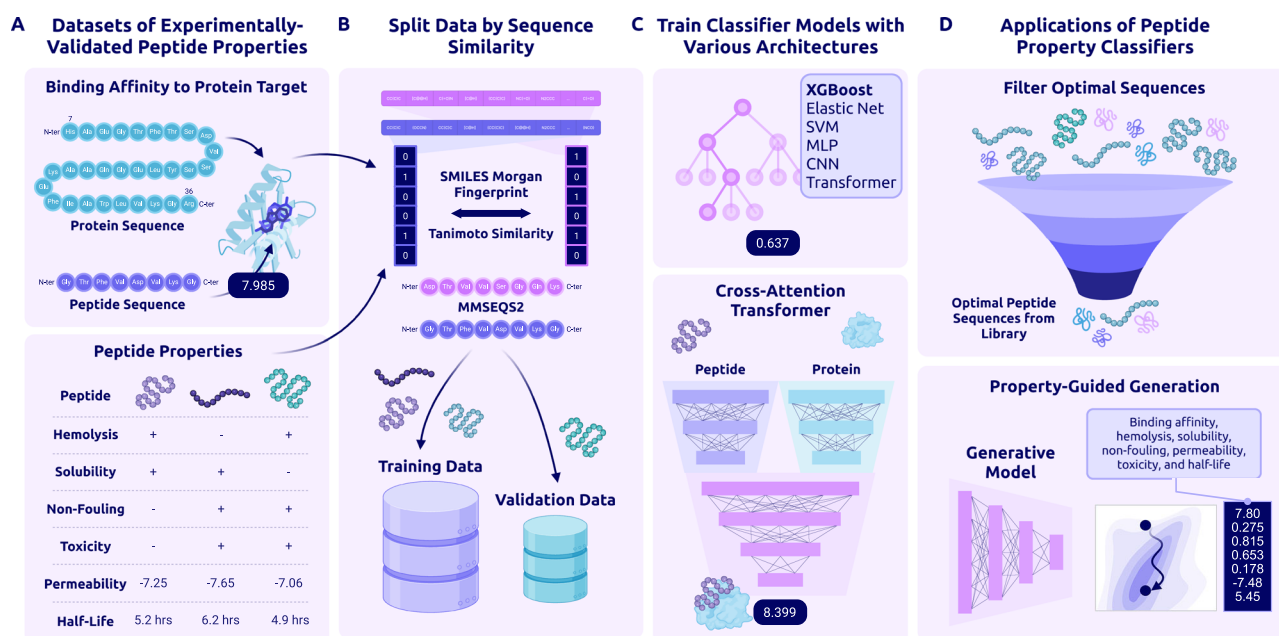


Fig. 1 | Overview of PeptiVerse workflow and applications. A Dataset collection: We collected datasets of experimentally-derived measurements for the following peptide properties: hemolysis, solubility, non-fouling, toxicity, permeability, half-life, and binding affinity to protein target(s). Measurements for all properties were collected across both canonical and non-canonical peptides. **B** Data splitting: Non-canonical peptides, represented by SMILES sequences, were split via Tanimoto similarity of Morgan fingerprints. Canonical peptide sequences were split via MMseqs2. **C** Property prediction: For most properties, we compared performance

across multiple architectures, including XGBoost, Elastic Net, support vector machine (SVM), multi-layer perceptron (MLP), convolutional neural network (CNN), and Transformer. The binding affinity regression model is a cross-attention transformer operating upon the peptide and protein sequence representations. **D** Applications: PeptiVerse predictions can be used to filter existing peptides for desired properties or as a reward signal for property-guided generative models. Created in BioRender. Chatterjee, P. (2026) <https://BioRender.com/h642117>.

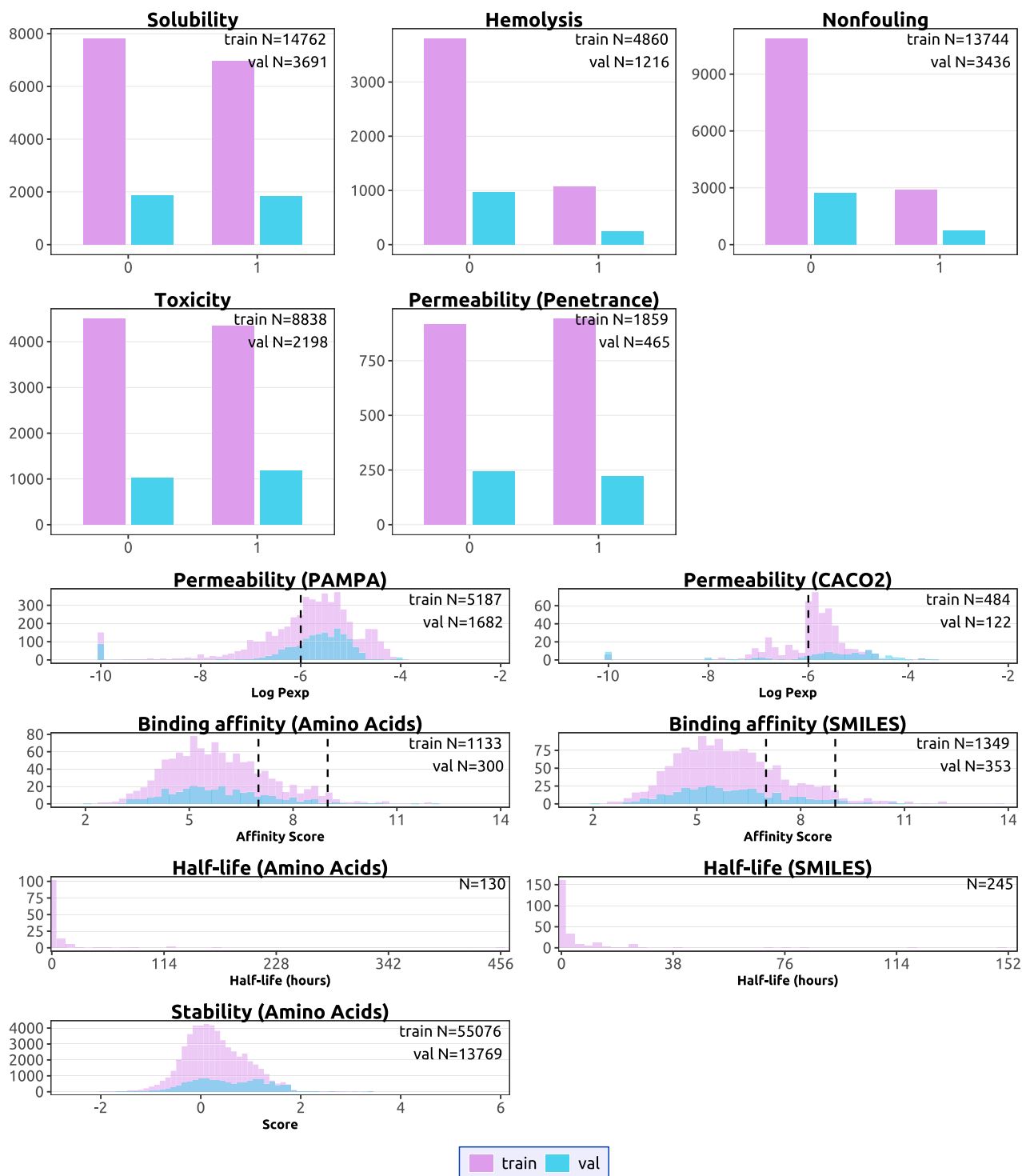


Fig. 2 | Training data distributions. Top: Distributions of binary-labeled datasets across training and validation splits. Bottom: Distributions of continuous-valued datasets across training and validation splits. Purple denotes training samples and blue denotes validation samples. Dashed vertical lines indicate heuristic thresholds

for favorable properties, including good permeability ($x = -6$) and strong binding affinity ($x = 7$ and 9). Due to limited data availability, half-life predictors were trained using cross-validation and therefore are shown without an explicit train/validation split. Source data are provided as a Source Data file.

alone, motivating property-specific modeling strategies and emphasizing the importance of continued dataset expansion.

PeptiVerse deploys state-of-the-art predictors for a broad range of therapeutically relevant peptide properties

Given our heterogeneous data settings, we evaluated a diverse set of predictor architectures within PeptiVerse, including linear models,

boosting methods, multilayer perceptrons (MLPs), convolutional neural networks (CNNs), support vector machines (SVMs), and transformer-based models (Fig. 3). Across all classification tasks and both amino acid sequence and SMILES inputs, overall performance differences between architectures were modest when trained on fixed, information-rich embeddings derived from ESM-2, PeptideCLM or ChemBERTa (Supplementary Fig. S1 and Supplementary

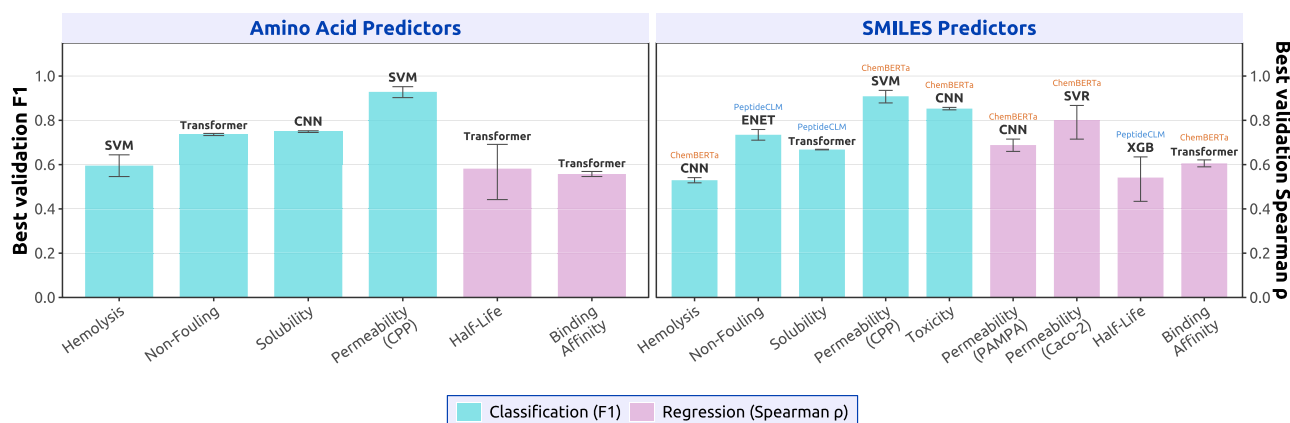


Fig. 3 | Best-model performance of PeptiVerse across different therapeutic peptide properties. Left: Performance on amino acid sequences encoded using ESM-2 embeddings; Right: Performance on SMILES sequences encoded using PeptideCLM and ChemBERTa embeddings. Bars report validation set performance of the best-performing model selected after OPTUNA optimization (200 trials), with the winning architecture annotated above each bar. For deep learning models (MLP, CNN, Transformer), the mean and error bars (95% CI) are derived from $N = 5$ independent training runs differing only in random seed (seeds: 1986, 42, 0, 123,

12345) and constitute technical replicates of the model training procedure. For non-ensemble models (XGBoost, SVR, ElasticNet), error bars represent 95% conformal prediction intervals from a single optimized model. Cyan indicates the performance of the classification models, and light purple indicates the performance of the regression models. For SMILES-based models, the embedding type (ChemBERTa or PeptideCLM) corresponding to the best-performing model is annotated in orange or blue, respectively, above the error bar. Source data are provided as a Source Data file.

Tables S1, S2, S3). This consistency suggests that representation quality and dataset characteristics, rather than downstream model capacity, constitute the primary performance bottleneck in peptide property prediction.

Rather than enforcing a single architecture, PeptiVerse identifies the best-performing model for each property and representation, reflecting the observation that no single architecture emerges as universally optimal. Different peptide properties favored different inductive biases, with hemolysis and permeability benefiting from margin-based or kernel methods, and chemically rich SMILES-based favoring convolutional architectures (Fig. 3). Importantly, similar trends were observed for regression tasks, where multiple nonlinear models achieved comparable performance, and no clear architectural dominance was observed (Supplementary Figs. S2, S3). In both settings, properties supported by larger datasets with broader coverage of the underlying value distributions, such as the cell-penetrating peptide permeability task (Permeability CPP) task, consistently yielded stronger performance than properties with limited or sparsely distributed data, such as peptide half-life. In addition, all evaluations were conducted using similarity-based data splits, indicating that simple models can generalize comparably to deep architectures on out-of-distribution peptide sequences when initialized with high-quality representations.

Across all tasks, embedding choice was the dominant factor governing predictive accuracy rather than model architecture. The spread in performance across embeddings consistently exceeded the spread across models within the same embedding. For regression, ChemBERTa consistently outperformed PeptideCLM on both PAMPA ($\rho = 0.69$ vs. 0.59) and Caco-2 ($\rho = 0.80$ vs. 0.75 ; Supplementary Table S2), likely reflecting ChemBERTa's broader pretraining chemical diversity compared to PeptideCLM's focus on synthetic cyclic peptides. DNN models have low variance with different random seeds, confirming that performance differences across configurations reflect genuine embedding and task effects rather than optimization noise. Although we find that baseline embeddings (e.g., ECFP, VHSE, or one-hot) can achieve competitive performance at times, embeddings from language models consistently attain the highest performance across properties (Supplementary Tables S4, S5, S6, S7, S8, S9). We find that ESM-C embeddings only improve upon ESM-2 embeddings for solubility ($F1 = 0.765$ vs. 0.754) and half-life ($\rho = 0.636$ vs. 0.582 ;

Supplementary Table S10), highlighting ESM-2's competitive representation ability.

PeptiVerse provides accurate binding affinity predictions

To assess whether PeptiVerse binding affinity predictions reflect physically meaningful interaction strength, we examined their relationship to structure-based confidence metrics derived from state-of-the-art de novo protein-peptide complex predictors. Recent studies have reported correlations between structure prediction confidence scores, such as ipTM, and docking-based interactions metrics for protein-protein interactions, motivating the use of such scores as proxies for binding strength in ab initio modeling pipelines^{52,53}. We therefore asked whether similar relationships hold for peptide-protein interactions, including both wild-type and chemically modified peptides.

Using OpenFold3-predicted complexes⁵⁴, we compared ipTM scores against experimentally measured binding affinities for 1433 amino acid sequence inputs and 1702 SMILES-based peptide inputs. In contrast to prior observations demonstrating the efficacy of the ipTM score for estimating protein-protein interaction strength⁵³, ipTM showed negligible association with experimental binding affinity across either peptide representation (Supplementary Fig. S4; $|\rho| \approx 0.1$), indicating that structure confidence metrics are insufficient proxies for peptide-protein binding strength. These results suggest that ipTM does not reliably capture the energetic or kinetic determinants governing peptide binding, likely due to the increased flexibility, shallow binding interfaces, and diverse chemistries characteristic of peptide ligands. Moreover, even when accurate complex structures are available, structure prediction is computationally heavier than embedding-based inference and is not well-suited for high-throughput screening across large peptide libraries. PeptiVerse therefore provides a fast affinity surrogate that complements structure prediction rather than replacing it.

PeptiVerse binding affinity predictions show statistically significant agreement with experimental measurements across both representation modalities, achieving Spearman $\rho = 0.57$ for amino acid sequence inputs and $\rho = 0.61$ for SMILES-based inputs (all $p < 10^{-3}$; Supplementary Fig. S3). These results, obtained using a transformer-based cross-attention architecture operating directly on protein and peptide embeddings, demonstrate that PeptiVerse provides experimentally relevant binding affinity estimates that complement

Table 1 | Results for property prediction benchmarks

Method	Regression (Spearman Correlation)			Classification (Best F1)	
	Binding Affinity (AA)	Permeability (all SMILES combined)	Binding Affinity (SMILES)	Permeability_CPP	Solubility
ESM-2	0.452	–	–	0.885	0.725
PepLand	0.503	0.628	0.768	0.838	0.662
PepLand + ESM-2	0.454	–	–	0.885	0.730
PeptiVerse	0.567	0.746	0.609	0.929	0.744

The results reported for PepLand are taken directly from the original publication²³, which uses random data splits, whereas PeptiVerse employs similarity-aware splits to evaluate out-of-distribution generalization. AA denotes amino acid sequence inputs. Results for PeptiVerse were reported for models at seed 1986. Source data are provided as a Source Data file. Bold text indicates the top score for each task.

structural modeling and enable binding-aware filtering and generative peptide design.

PeptiVerse demonstrates superior predictive performance relative to existing peptide property predictors

PeptiVerse is, to our knowledge, the first unified framework capable of predicting multiple physicochemical and developability properties for both amino acid sequence and SMILES-encoded peptide inputs. In contrast, prior tools have made meaningful progress but remain specialized in either input modality or property scope^{23,27}. For example, PeptideBERT¹⁵ operates primarily on canonical amino acid sequences and does not generalize to chemically modified peptides, whereas PepLand²³ handles SMILES-based representations of both canonical and non-canonical peptides but covers only a limited set of properties. None of the existing tools provides the breadth of properties, multimodality, or user accessibility offered by PeptiVerse.

To quantitatively assess these differences, we compare PeptiVerse against representative prior methods across multiple regression and classification benchmarks in Table 1. Across most evaluated tasks, PeptiVerse achieves competitive or superior performance under a unified evaluation protocol. We note that PepLand reports higher performance on SMILES-based binding affinity prediction²³. However, these results were obtained using random data splits, which permit substantial overlap in sequence or chemical similarity between the training and test sets. In contrast, for binding affinity PeptiVerse employs score-distribution-based splits, which ensure both splits span the full affinity range, as targets and binders recur across many peptide-protein pairs and cannot be cleanly separated by similarity. Performance differences on SMILES-based binding affinity thus reflect differences in the evaluation protocol rather than representational limitations.

Additionally, the PepLand + ESM-2 setting is reported only for canonical peptide tasks, and the mechanism for switching between SMILES and sequence inputs is not explicitly specified. By contrast, PeptiVerse provides an explicit and unified multimodal design, enabling consistent evaluation across both canonical and non-canonical peptide spaces.

PeptiVerse is deployed as a unified web interface for peptide evaluation

Finally, we developed an interactive web interface to make PeptiVerse accessible for practical use by both experimental and computational researchers. PeptiVerse is deployed as a web server (Fig. 4) built with Gradio⁵⁵ and hosted on HuggingFace Spaces⁵⁶. The interface allows users to submit peptide inputs as either amino acid sequences or SMILES strings, enabling property evaluation across both canonical and chemically modified peptides. It supports a broad set of therapeutically relevant properties, including binding affinity to a target protein sequence, hemolytic activity, non-fouling behavior, permeability, solubility, toxicity, and peptide half-life.

To improve interpretability and usability, the interface also provides visualizations of the underlying training data distributions,

allowing users to contextualize input peptides relative to experimentally characterized datasets. All datasets used to train the deployed models are standardized and distributed in HuggingFace Dataset format, facilitating reproducibility, benchmarking, and integration into downstream peptide design and optimization workflows.

Discussion

PeptiVerse introduces a unified framework for therapeutic peptide property prediction that supports both amino acid sequence and SMILES-encoded inputs, including peptides containing non-canonical amino acids. By building on pretrained protein (ESM-2) and chemical (PeptideCLM) language models^{26,27}, PeptiVerse focuses on training lightweight predictor heads rather than full representation models, enabling efficient, scalable, and easily deployable property evaluation. These results highlight the heterogeneity of peptide property landscapes and motivate a flexible, property-aware modeling strategy rather than reliance on a single predictor.

Compared with prior SMILES-based peptide models such as PepLand²³, which rely on complex graph constructions and full-model retraining, PeptiVerse shows that language model embeddings paired with simple, well-regularized classifiers are sufficient (and often superior) for practical peptide property prediction. This design reduces computational cost while improving generalizability, making PeptiVerse compatible with modern peptide design pipelines. The advantage is particularly clear for binding affinity, where structure prediction confidence alone fails to track peptide-protein binding strength (Supplementary Fig. S4), whereas PeptiVerse yields strong, statistically significant agreement with experimental measurements (Supplementary Table S5). Together, these results motivate fast, data-driven affinity predictors that complement rather than replace structural modeling, and position PeptiVerse as an open, extensible benchmark for therapeutic peptide discovery.

Importantly, PeptiVerse is designed as a lightweight and accessible framework that supports direct incorporation of property predictors into iterative and high-throughput design workflows, rather than limiting use to web-based post hoc evaluation⁵⁷. This design enables a natural application to generative peptide sequence modeling, where efficient property evaluation is required to guide optimization. In this setting, PeptiVerse predictors enable rapid assessment of therapeutically relevant properties, allowing generative models to produce sequences with a higher likelihood of translating in wet-lab settings^{29,30,32–34}. Such guidance relies on fast reward evaluation to support gradient-based or iterative refinement over generated sequences⁵⁸, particularly in multi-objective settings where properties such as binding affinity, solubility, and toxicity must be jointly optimized.

In practice, PeptiVerse predictors have already been integrated with state-of-the-art generative frameworks, including PepTune³³, which introduces Monte Carlo Tree Guidance for multi-objective-guided peptide optimization, TR2-D2³⁴, which applies reinforcement learning for reward fine-tuning of peptide discrete diffusion models, and moPPit²⁸, which enables motif-specific peptide generation via

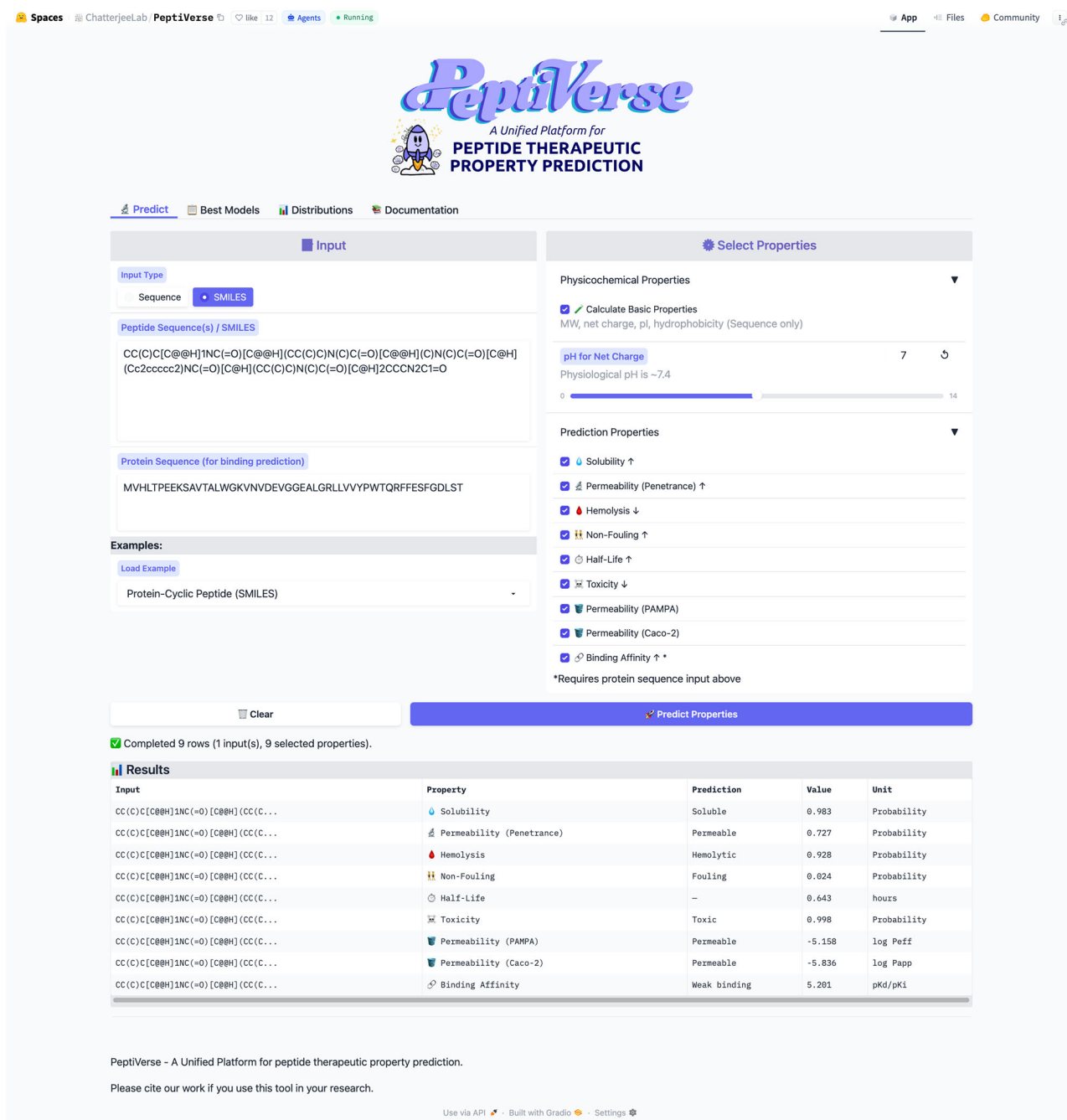


Fig. 4 | Web interface of PeptiVerse hosted on HuggingFace. The interface allows users to submit peptide inputs as either amino acid sequences or SMILES strings and select from a range of physicochemical and predictive properties for evaluation. Property options are dynamically enabled based on input type, and basic

properties such as molecular weight, charge, and hydrophobicity can be computed alongside learned predictors. This design enables rapid, no-code peptide property assessment suitable for both exploratory analysis and integration into iterative design workflows.

discrete flow matching and is experimentally verified across multiple target classes. Together, these examples position PeptiVerse as a modular reward evaluation layer that integrates naturally into modern generative pipelines, supporting both inference-time steering and training-time optimization.

To ensure continued relevance, PeptiVerse will be updated regularly as new data and predictors become available. While several property models already achieve strong performance, others (i.e., chemically-modified peptide half-life in SMILES space) remain constrained by limited and heterogeneous experimental data. Importantly, the modular design of PeptiVerse enables seamless integration of improved representation models as they emerge; for example,

retraining with newer embeddings such as ESM-C⁵⁹ provides performance gains over ESM-2 on properties such as half-life and solubility (Supplementary Tables S8 and S10), further supporting that advances in representation learning directly translate to improved predictive accuracy within this framework. We therefore encourage open-source deposition of peptide property measurements from both academia and industry. In parallel, we are developing new specialized binding predictors, including peptide isoform-specificity³⁶, motif-specificity²⁸, and metal-binding propensity³⁵, which will be incorporated as additional properties within the PeptiVerse framework. By providing a standardized, extensible, and openly accessible platform for integrating new data and models, PeptiVerse establishes a practical

mechanism for expanding predictive scope, enabling property-guided generative design, and supporting the translation of next-generation peptide therapeutics.

Methods

Data collection and preparation

Throughout this work, we distinguish peptide inputs by representation modality rather than biological origin. “Amino acid” inputs refer to canonical sequence-based representations processed by protein language models, while “SMILES” inputs refer to chemistry-aware molecular representations, which may include both canonical and non-canonical peptides.

Hemolysis. Hemolysis data were retrieved from PeptideBert and peptide-dashboard^{15,25}, and cross-validated against the original experimental records in DBAASP v3.0⁴¹. Peptides labeled as 1 are considered hemolytic, whereas 0 denotes non-hemolytic activity. The final dataset comprised 4765 non-hemolytic and 1311 hemolytic peptide entries.

Permeability. All permeability annotations were obtained from PepLand²³, which sources experimental measurements from CycPeptMPDB¹⁶. The noncanonical dataset contains 7475 non-canonical peptides with reported permeability values measured using either PAMPA⁴⁹ or Caco-2 assays⁵⁰. Permeability is reported as $\log P_{\text{exp}}$, the logarithm of the effective permeability coefficient, which reflects the peptide’s lipophilicity and ability to passively diffuse across lipid membranes.

PAMPA and Caco-2 assays quantify different biological processes, with PAMPA measuring passive membrane permeability and Caco-2 assays relating more closely to intestinal absorption potential^{49,60,61}. For this reason, the two assay types were handled separately during data preparation, not following PepLand’s combined training strategy. There were 6,869 PAMPA sequences and 606 Caco-2 sequences collected in the end. Following CycPeptMPDB conventions, peptides with $\log P_{\text{exp}} \geq -6.0$ were labeled as high permeability, and the remaining peptides as weak permeability¹⁶. The predictive tasks associated with these datasets are called Permeability (PAMPA) and Permeability (Caco-2).

The canonical permeability dataset contains 1162 cell-penetrating peptides and 1162 non-penetrating peptides, curated from PepLand²³. This dataset was constructed to balance peptide length distributions between positive and negative classes. Positive peptides were originally collected from 22 independent cell-penetrating peptide studies, while negative examples were sourced from UniProt. A notable fraction of the positive peptides (~8.7%) exhibited low sequence complexity, defined as a ratio of peptide length to the number of unique amino acids greater than 5, compared to only 0.95% among non-penetrating peptides. Although these low-complexity sequences are likely engineered, they were retained, as they may encode informative features relevant to membrane permeability²³. The predictive task associated with this dataset is called Permeability_CPP.

Non-fouling. Non-fouling annotations were obtained from PeptideBERT and Peptide-Dashboard^{15,25}, both of which source data from the dataset of Barrett et al.⁴⁷. In this context, non-fouling peptides are defined as sequences that resist nonspecific protein adsorption, while fouling peptides permit such adsorption, which can lead to functional loss and reduced performance^{47,62}. Peptides labeled as 1 are classified as non-fouling, and those labeled as 0 are considered fouling. The curated dataset comprised 13,580 fouling and 3600 non-fouling peptide entries.

Toxicity. Toxicity data were obtained from ToxinPred3.0, which provides canonical amino acid peptide sequences with experimentally validated toxicity labels⁴². The dataset contains 5518 toxic and 5518

non-toxic peptides, where label 1 denotes toxic, and 0 denotes non-toxic. Amino acid sequences were converted into SMILES format with the `fasta2smi` command from `p2smi`⁶³. Molecular redundancy was reduced by clustering peptides using Morgan fingerprints (radius 2, 2048 bits, including chirality) and RDKit’s Butina clustering algorithm with a similarity threshold of 0.6.

Solubility. Solubility labels follow the protocol established in PROSO II⁴³, as used in PeptideBERT¹⁵, with additional sequences incorporated from SoluProtMutDB⁴⁸. In PROSO II, the soluble class (1) was assigned based on experimental metadata recorded in pepcDB, following the Protein Structure Initiative (PSI) pipelines⁶⁴. A sequence was labeled soluble once it reached the “Soluble” stage or any downstream stage. Additional soluble entries were derived from PDB records (up to 2010) annotated with expression from *E. coli*, as crystallographic and NMR structure determination requires a protein that has been purified in solution. The insoluble class (0) consisted of constructs that remained in the “not Soluble” state for at least eight months, based on the different pepcDB releases. This yields a combined dataset of 8,785 soluble and 9668 non-soluble sequences in total.

Binding affinity. Binding affinity prediction was performed using paired protein sequences and peptide SMILES from the PepLand dataset²³, which aggregates different experimental measurements together (K_d , K_i , IC_{50}). Their data collection protocol follows CAMP⁵¹, which subsets data from RCSB PDB⁶⁵ and DrugBank entries containing the label of “peptide”⁶⁶. All scores were transformed based on the negative logarithm of the original affinity data into a unified scale. A shared unit scale was roughly designed to indicate the strength of binding: with 9 indicating strong *nM* to *pM* binders, 7–9 indicating *nM* to μM medium binders, and < 7 indicating weak μM binders. The data split was based on affinity score distribution matching, ensuring both splits contain a similar distribution of data. The final dataset contained 1433 peptide-protein pairs with canonical peptide sequences (amino acid inputs) and 1702 pairs with SMILES-encoded peptides containing non-canonical amino acids.

$$\text{affinity_score} = -\log_{10}(K_{d/i} \text{ in M}) \quad (1)$$

Thus, higher values indicate stronger binding, which is more intuitive for modeling.

Stability. Stability data for amino acid sequence inputs were collected and organized from TAPE⁶⁷ and SaProt⁶⁸ (Huggingface: SaProtHub/Dataset-Stability-TAPE) and used to pretrain stability predictors before half-life modeling. Stability was treated as a continuous score reflecting the ability of proteins to remain folded above concentration thresholds. A total of 68,845 sequences were collected.

Half-life. Half-life data were compiled from THPdb2⁴⁴, PEPLife⁴⁵, and PepTherDia⁴⁶. Only human serum measurements were retained. Reported values across datasets vary in units and granularity. All half-life measurements were therefore converted into hours for interpretability. Details of the data construction are provided in the Supplementary Information. The curated dataset comprised 130 amino acid sequence entries and 245 SMILES-based sequences.

Protein-peptide ipTM scores. The interface predicted TM-score (ipTM) is a confidence metric originally designed to assess the accuracy of predicted protein-protein interaction interfaces in multimeric structure prediction models⁶⁹. All ipTM scores for protein-peptide pairs, including SMILES-encoded pairs, were obtained from OpenFold3³⁴ via its API on the NVIDIA NGC platform⁷⁰. MSA profiles for the target proteins were constructed via MMseqs2⁷¹ against the UniRef30 database⁷².

Table 2 | Dataset composition across peptide property prediction tasks

Properties	Amino Acid Sequences		SMILES Sequences	
	0	1	0	1
Classification				
Hemolysis	4765	1311	4765	1311
Non-Fouling	13580	3600	13580	3600
Solubility	9668	8785	9668	8785
Permeability_CPP	1162	1162	1162	1162
Toxicity	–	–	5518	5518
Regression (N)				
Permeability (PAMPA)	–	–	6869	
Permeability (CACO2)	–	–	606	
Half-Life	130		245	
Binding Affinity	1433		1702	

Classification tasks report the number of samples per class (0/1), where 0 is negative and 1 is positive. Regression tasks report the total number of samples (N). Counts are shown separately for amino acid and SMILES sequence representations, illustrating differences in data availability and class balance across properties. Source data are provided as a Source Data file.

Additional properties. Additional physicochemical features, including isoelectric point, molecular weight, and hydrophobicity, were computed using utilities from the Biopython package⁷³. These properties were calculated dynamically with a tunable pH parameter to account for protonation-state dependence.

Data splitting. All properties defined on amino acid-based peptide representations were split using a shared clustering strategy. Sequences from all amino acid sequence datasets were clustered independently using MMseqs2⁷¹ with identical parameters (`–min-seq-id 0.3 –c 0.8 –cov-mode 0`) and split using an 80/20 cluster-level split to prevent sequence leakage between training and evaluation subsets. Amino acid sequences were converted into SMILES format with the `fasta2smi` command from `p2smi`⁶³. For properties evaluated using both sequence and SMILES inputs (hemolysis, non-fouling, solubility, permeability_CPP), the original cluster-based train/validation assignments were preserved after conversion, ensuring fair comparison across input modalities. For datasets natively represented in SMILES notation, molecules were clustered using Morgan fingerprints and Tanimoto distance via RDKit⁷⁴, followed by an 80/20 cluster-level split. Datasets used for binding affinity prediction, which contain paired peptide-protein inputs across two molecular modalities, were instead split by matching affinity score distributions between training and evaluation sets. Detailed dataset sizes, label distributions, and train/validation splits for all properties are summarized in Table 2 and Fig. 2.

Model architecture and training

Sequence and SMILES representations. Protein sequences were represented using embeddings from ESM-2 (`esm2_t33_650M_UR50D`)²⁶, while peptide sequences were represented using either ESM2, PeptideCLM (PeptideCLM-23M-all)²⁷, which employs a tokenizer of size 586 designed to capture noncanonical peptide chemistry, or ChemBERTa (ChemBERTa-77M-MLM)⁷⁵, trained across millions of chemical structures curated from PubChem. See Supplementary Methods for details on other embedding approaches, including ESM-C and baseline embeddings. Depending on the downstream model, embeddings were either average-pooled across sequence positions to form fixed-length representations or retained in unpooled, position-resolved form to preserve residue-level and positional information. This distinction allowed different model classes to exploit either global sequence summaries or fine-grained positional structure.

Predictor architectures. For properties including toxicity, solubility, hemolysis, non-fouling, and permeability, lightweight predictor heads were trained on frozen foundational embeddings. We evaluated a broad set of predictor architectures, including multilayer perceptrons (MLPs), convolutional neural networks (CNNs), and transformer-based models, as well as classical statistical learners such as Elastic Nets (ENET)^{38,76}, support vector machines (SVMs)³⁷, epsilon-support vector regression (SVR)⁷⁷, and XGBoost (XGB)³⁹. ENET and SVM classifiers were implemented using the RAPIDS cuML library⁷⁸ to enable GPU acceleration, while SVR models were implemented using `scikit-learn`⁷⁹. Classification models were trained using standard objective functions appropriate to each method, whereas regression models were optimized using mean squared error. All models were trained using their canonical formulations without architectural modification. For CNN- and transformer-based predictors, unpooled embeddings were used to allow convolutional kernels and self-attention mechanisms to explicitly exploit positional structure, while pooled embeddings were used for MLP and tree-based models operating on fixed-dimensional inputs.

Hyperparameter optimization. Hyperparameters for all models were optimized using the Optuna framework⁸⁰, with an initial target of 200 optimization trials per configuration (Supplementary Tables S12–S15). For parameters sampled on a logarithmic scale, values were drawn uniformly in log space to efficiently explore multiple orders of magnitude. For regression tasks, both mean squared error (MSE) and Huber loss were evaluated, using the latter to improve robustness to outliers via its δ parameter controlling the transition between L1 and L2 penalties. For computationally intensive configurations exceeding approximately 10 hours of wall-clock training time, the number of Optuna trials was reduced to 50 or 20 while preserving representative coverage of the hyperparameter space. Results of the hyperparameter exploration are reported in Supplementary Figs. S2 and S5.

Binding affinity prediction. Binding affinity prediction was formulated as a regression task using a transformer-based architecture with cross multi-head attention to learn a joint latent representation between peptide and protein modalities. The model architecture was fixed across experiments, with variation introduced only through peptide embedding initialization. Peptide inputs were represented using either ESM-2 or PeptideCLM embeddings^{26,27}, in pooled or unpooled form, while protein targets were consistently represented using ESM-2 embeddings. The model produced a continuous affinity score, which was additionally mapped to three discrete affinity classes, yielding a multitask framework aligned with other property predictors. Hyperparameters were optimized using 200 Optuna trials, and Spearman's rank correlation coefficient (ρ) was used as the primary model selection criterion (Supplementary Table S15 and Supplementary Fig. S2).

Half-life prediction. Peptide half-life prediction was formulated as a regression task using both amino acid sequence inputs and SMILES-based representations. For amino acid-based inputs, models were initialized from pre-trained stability predictors trained on the wild-type stability dataset and subsequently fine-tuned on half-life data. The same model families used for binary property prediction were retained. Four regression configurations were evaluated: (i) XGBoost predicting half-life in hours, (ii) XGBoost predicting $\log(1 + \text{half} - \text{life})$, (iii) a transformer-based regressor predicting half-life in hours, and (iv) a transformer-based regressor predicting $\log(1 + \text{half} - \text{life})$. Hyperparameters were optimized using Optuna with five-fold cross-validation, using 200 trials for XGB, SVR, and ENET models and 50 trials for CNN, MLP, and Transformer models, with 20 training epochs per trial.

The transformation equation is defined as follows:

$$\text{half} - \text{life_score} = \ln(1 + t_{1/2}) \quad (2)$$

where $t_{1/2}$ is the half-life in hours.

For SMILES-based half-life prediction, models were trained directly on peptide SMILES embeddings without intermediate stability pretraining. Predictor architectures matched those used for amino acid-based half-life prediction and were trained to predict $\log(1 + \text{half} - \text{life})$. Evaluation was performed using five-fold cross-validation. Optuna optimization employed 200 trials for XGB, SVR, and ENET models and 50 trials for CNN, MLP, and Transformer models, with transformer-based models trained for 100 epochs per trial.

Evaluation metrics. Model evaluation metrics were selected according to task type. Classification performance was assessed using F1 score and area under the receiver operating characteristic curve (AUC) with results reported in Supplementary Fig. S1 and Supplementary Table S3. Regression performance was evaluated using Pearson correlation coefficient (r), Spearman's rank correlation coefficient (ρ), and the coefficient of determination (R^2), with results summarized in Supplementary Fig. S3.

Model stability, confidence intervals, and uncertainty estimation. All deep learning models (MLP, CNN, Transformer) were retrained five times with distinct random seeds (1986, 42, 0, 123, 12345) using fixed optimal hyperparameters from OPTUNA. The 95% confidence intervals for classification (F1, AUC) and regression (Spearman ρ , RMSE, R^2) metrics were estimated as $\bar{x} \pm t_{0.025, 4} \cdot \frac{s}{\sqrt{5}}$. Per-sequence uncertainty was quantified as the standard deviation of predictions across seeds.

For classical ML models (SVM, ElasticNet), optimization is convex and therefore deterministic given fixed data and hyperparameters⁸¹, leading to negligible variability across random seeds. For XGBoost, while stochastic subsampling of features and instances introduces minor seed dependence³⁹, empirical variability is small relative to data sampling variability. Confidence intervals for all three model types were therefore derived via non-parametric percentile bootstrap ($n = 2000$) over held-out validation predictions ($y_i, \hat{y}_i$)⁸² without retraining. The percentile method was preferred over symmetric t -intervals to respect the bounded range of Spearman $\rho \in [-1, 1]$, with Fisher z -transform intervals computed as a cross-check when $\rho > 0.9$. The decision threshold was fixed on the full validation set and held constant across bootstrap resamples to avoid optimistic bias. Per-sequence uncertainty was estimated as the probability margin from the decision boundary for probabilistic classifiers, including XGBoost ($c_i = 2|p_i - 0.5|$), and as empirical residual intervals ($\hat{y}_i \pm z_{0.975} \hat{\sigma}$) for deterministic regression models (SVR, ElasticNet).

For the half-life regression models, 95% confidence intervals were computed by non-parametric percentile bootstrap over out-of-fold prediction pairs ($y_i, \hat{y}_i$) obtained from cross-validation without retraining. For each bootstrap resample, RMSE, MAE, R^2 , and Spearman's ρ were recomputed, and the 2.5th and 97.5th percentiles of the resulting empirical distributions were reported as the interval bounds. In addition to the point estimate on the full validation set, we also report the bootstrap mean and standard deviation for each metric.

Ablation of full fine-tuning and LoRA for half-life prediction. To further assess the role of parameter-efficient adaptation in this data-limited setting, we performed an additional half-life transfer-learning ablation comparing full fine-tuning and low-rank adaptation (LoRA) using the stability-pretrained transformer⁸³ (Supplementary Table S11). This comparison was restricted to the transformer architecture, as it was the strongest sequence-based stability model

and the most appropriate setting for LoRA-based adaptation. In the LoRA setting, the pretrained stability checkpoint was first loaded, selected linear layers in the transformer were replaced with LoRA-augmented modules while the backbone weights were frozen, and only the low-rank update matrices and the final regression head were trained. Performance was evaluated for both raw and log-transformed half-life targets. Full fine-tuning outperformed LoRA under both target formulations, with the best results obtained for full fine-tuning on log-transformed half-life, whereas LoRA remained competitive but consistently lower-performing. These results suggest that transfer from stability prediction to half-life prediction benefits from broader adaptation of the pretrained representation, while log-space prediction provides a more favorable learning target for this task.

Inference-time confidence estimation. To provide confidence scores per-prediction at inference time, different strategies were applied according to the model class, following the framework of Lakshminarayanan et al.⁸⁴.

For classification tasks, uncertainty is quantified as the predictive entropy of the model's output probability. For deep learning classifiers (MLP, CNN, Transformer), inference-time uncertainty was quantified using the deep ensemble of five independently seeded models. Given a query input $\mathbf{x}$, each model $m \in \{1, \dots, 5\}$ produces output probability $p_{\theta_m}(y|\mathbf{x})$. The ensemble predictive distribution is the uniformly-weighted mixture $\bar{p}(y|\mathbf{x}) = \frac{1}{5} \sum_{m=1}^5 p_{\theta_m}(y|\mathbf{x})$. For classical classifiers (XGBoost, SVM), $\bar{p}(y|\mathbf{x})$ is the single model's output probability. For linear classifiers (ElasticNet) that produce a decision function score $f(\mathbf{x})$ rather than a native probability, $\bar{p}(y|\mathbf{x}) = \sigma(f(\mathbf{x}))$ where σ denotes the sigmoid function. In both cases, per-sample uncertainty is quantified as the binary predictive entropy:

$$\mathcal{H}[\bar{p}] = -\bar{p} \log \bar{p} - (1 - \bar{p}) \log(1 - \bar{p}) \quad (3)$$

High entropy indicates either ensemble disagreement (epistemic uncertainty) or collectively diffuse predictions (aleatoric uncertainty). The deep ensemble formulation is preferred over MC Dropout⁸⁵ as deep ensembles explore multiple modes of the loss landscape via independent random initialization, whereas MC Dropout samples within a single basin, leading to systematically overconfident predictions on out-of-distribution inputs⁸⁴.

For regression tasks, uncertainty estimation depends on the model class. For deep learning regression models (MLP, CNN, Transformer), inference-time uncertainty is reported as the standard deviation of point predictions across the five ensemble members:

$$\sigma_{\text{ens}}(\mathbf{x}) = \sqrt{\frac{1}{5} \sum_{m=1}^5 (\hat{y}_m(\mathbf{x}) - \bar{y}(\mathbf{x}))^2} \quad (4)$$

where $\bar{y}(\mathbf{x}) = \frac{1}{5} \sum_{m=1}^5 \hat{y}_m(\mathbf{x})$ is the ensemble mean prediction. For deterministic classical regression models (SVR, ElasticNet) where seed-based ensembling is inapplicable due to convex optimization yielding identical solutions, and for XGBoost regression, where retraining seed ensembles was outside the scope of this work, inference-time prediction intervals were constructed using the residual normalized conformity score^{86,87}. Rather than using MAPIE's sklearn API directly, we adopt a custom implementation since our base models embed raw sequences via GPU-accelerated transformer forward passes which are incompatible with sklearn's `fit(X, y)` interface, the binding affinity model requires dual sequence inputs, and held-out predictions were already precomputed during training. An auxiliary XGBoost residual estimator $\hat{\sigma}$, that provides stronger nonlinear modeling than the linear default in MAPIE, is fitted on precomputed held-out embeddings and absolute residuals $|y_i - \hat{y}_i|$. Normalized conformity scores $s_i = |y_i -$

$\hat{y}_i/\hat{\sigma}(\mathbf{x}_i)$ are then computed, and the conformal quantile defined as:

$$q = \text{Quantile} \left(\{s_i\}_{i=1}^n, \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right) \quad (5)$$

is taken at the finite-sample corrected level, guaranteeing marginal coverage $\geq 1 - \alpha = 0.9$ under exchangeability⁸⁸. At inference, the prediction interval $[\hat{y}(\mathbf{x}) - q \cdot \hat{\sigma}(\mathbf{x}), \hat{y}(\mathbf{x}) + q \cdot \hat{\sigma}(\mathbf{x})]$ has width that varies per input, since $\hat{\sigma}(\mathbf{x})$ captures local prediction difficulty.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All processed datasets used to train PeptiVerse predictors are available at https://huggingface.co/datasets/ChatterjeeLab/PeptiVerse_data. All benchmarking model weights are available in Zenodo: <https://zenodo.org/records/19989009> and the best model weights are available at <https://huggingface.co/ChatterjeeLab/PeptiVerse>. Additional figures and tables are provided in a Supplementary Information file. Source data for all figure plots and tables, in both the main text and Supplementary Information file, are provided with this paper in the Source Data file. Source data are provided in this paper.

Code availability

The full set of predictors is available through a simple API at <https://huggingface.co/ChatterjeeLab/PeptiVerse>, which can also be used to download datasets. Code for plotting in R is available at https://github.com/ynuozhang/Peptiverse_R.git. For users who prefer a no-code interface, all predictors can also be easily run via an interactive HuggingFace Space at <https://huggingface.co/spaces/ChatterjeeLab/PeptiVerse>. All code is open source, with predictors and associated resources updated and maintained regularly.

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Author contributions

Y.Z., S.T., T.C., E.M. and S.V. trained models, curated datasets, and prepared results. Y.Z. managed dataset and model curation and wrote the paper, with input from S.T. and T.C. P.C. conceived, designed, and directed the study, and reviewed and finalized the manuscript.

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Competing interests

P.C. is a co-founder of Gameto, Inc., UbiquiTx, Inc., AtomBioworks, Inc., and Recognition Bio., Inc., and advises companies involved in peptide therapeutics development. P.C.'s interests are reviewed and managed by the University of Pennsylvania in accordance with their competing-of-interest policies. The remaining authors have no competing interests to declare.

Additional information

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